ANALYSIS OF AIRPORT SECURITY WITH QUEUING THEORY

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ABSTRACT. This paper presents two queuing models to simulate airport security gates while accounting for two types of passengers: PreCheck and regular. We use public data from the Transportation Security Administration (TSA) to estimate model parameters. For cases with no data, we back out parameter values using Erlang's C formula and Little's Law. Our first model is an M/M/c queue with pooled servers and a priority queue. Using the M/M/c model, we present the airport security's total cost function and find optimal service rates and number of servers (μ and c, respectively). We then relax the distributional assumptions of the M/M/c model to build an M/N/c model to simulate results. Using the M/N/c model, we model cost from the passenger perspective, finding the optimal PreCheck price based on the average wait time. We conclude with a sensitivity analysis of our results to the most important parameters.

Keywords. Queuing Theory, Optimization, Mathematical Modeling, Operations Research, Markov Chains
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1. Introduction

Airport security has a reputation for unpredictable wait times and for being confusing to navigate, but it is vital in ensuring the safety of travelers. It is a complex balancing act of minimizing passenger wait times, minimizing operational costs without compromising safety. Such an important system has motivated many research projects in attempts to find more optimal methods to decrease wait time. Most prior research relies heavily on continuous-time Markov chains [7], i.e., queuing models, to model security checkpoints, which we also do in this paper.

Security checkpoints have strict standards, which include luggage screening devices, conveyors, a metal detector, and additional tools for when a passenger is flagged [4]. Standard procedure includes the removal of metal accessories, exposing large electronics to individual scans, and removing shoes for inspection. TSA segments passengers into two groups: PreCheck and regular screening. PreCheck is a priority status for pre-screened passengers to reduce the steps needed to get through security. These include the privileges of keeping shoes on, leaving large electronics in luggage, and often a separate security line.

From a modeling perspective, a security checkpoint is a system with passengers in a queue and a pool of servers. Passenger interarrival times and service rates are random variables, with some underlying distribution. For the remainder of our project, the service rate encompasses all parts of a screen: checking ID, luggage screening, and walking through a metal detector. We do not model the case where an individual rejoins a queue due to a security failure. This paper also accounts for cost, from the airport and passenger perspective. In the former, TSA agents need to be paid, and slow service incurs an opportunity cost through lost customer wages because customers can get to the airport later if the wait time is shorter. In section 3.1, we present a total cost function to model a checkpoint's operational costs and perform nonlinear optimization to find the optimal number of servers and service rate (c^*, μ^*, respectively). Section 3.2 optimizes the cost from the passenger perspective by finding the optimal PreCheck price.

2. Description of the Model

In many prior studies, authors use parallel M/M/c queues. In this paper, we use the term parallel to mean separate lines, which can be interpreted as having independent queues. These models are often preferred because they have closed-form solutions, which can be studied with formal, mathematical analysis. Here, we propose modifications to the standard M/M/c models. The first is a priority queue model, where servers are pooled and PreCheck customers take priority. The second is an M/G/c model, where G denotes any distribution. For the remainder of the project, we will use a normal distribution, M/N/c. The motivation for M/N/c is that the exponential distribution assumption for interarrivals and service times is overly strong, especially the latter. Service times most likely aren't short with a rapid decay. We make the case for why service time should be normal in section 3.2. Note that neither the priority queue nor the M/N/c model has closed-form solutions. We rely on simulation for our results.

Just like a standard M/M/c model, our priority queue model treats arrivals as a single queue governed by a Poisson process with arrival rate λ. Interarrival times are exponentially distributed with rate parameter λ and mean interarrival time of $\frac{1}{\lambda}$. Mathematically, we can express the pdf of interarrival times as:

$$a(t) = \begin{cases} \lambda e^{-\lambda t} & \text{if } t \geq 0 \\ 0 & \text{if } t < 0 \end{cases}. \tag{2.1}$$

Service times are also assumed to be exponential. The pdf for service time is the same as above, but with μ denoting the service rate and mean service time of $\frac{1}{\mu}$. Finally, there are c total servers.

The value for λ was estimated using LAX passenger data from Tom Bradley International Terminal, with dates ranging from 1-1-2025 to 7-21-2025. The data provided was in units of passengers per hour. We convert the data to units of passengers per minute and take the average, since the sample average is the MLE estimate for a Poisson distribution. We use the same estimated λ as the parameter for the interarrival distribution. We arrive at $\lambda = 23.6$ passengers per minute [6].

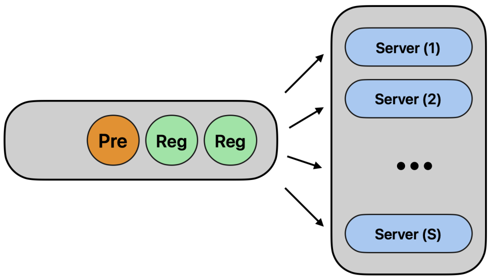


Figure 1. M/M/c Priority Queue

Estimating the service rate μ is more complex. The data only contains the arrival rate and the average wait time, not the service rate. We were unable to obtain data or find papers that properly estimated the service rate for regular queues and PreCheck queues. To circumvent the issue, we assume the data was generated from an M/M/c queue. Under this assumption, we can solve an optimization problem to find μ.

Let server utilization be $\rho = \frac{\lambda}{c\mu}$. The average number of customers in the queue can be written as,

$$L(\mu, \lambda, c) = L_q + L_s = \frac{\lambda}{c\mu - \lambda} \cdot C(c, \frac{\lambda}{\mu}) + \frac{\lambda}{\mu}. \tag{2.2}$$

In the first equality, we decomposed the wait time into two terms. L_q is the average number of customers in the queue, and L_s is the average number of customers in service. In the second equality, we expressed the equation in terms of Erlang's C Formula, $C(c, \lambda/\mu)$, which is the probability that an arriving customer is forced to join the queue [14]:

$$C(c,\rho) = \frac{(\frac{(c\rho)^c}{c!})(\frac{1}{1-\rho})}{\sum_{k=0}^{c-1} \frac{(c\rho)^k}{k!} + \frac{(c\rho)^c}{c!(1-\rho)}}. \tag{2.3}$$

Next, we invoke Little's Law [15], which is a conservation equation. The theorem states that the average number in the system is equal to the product of the arrival rate and average system time [5].

$$L_q = \lambda W_q \tag{2.4}$$

We now have an expression for the expected waiting time W_q:

$$W_q = \frac{L_q}{\lambda} = \frac{C(c,\rho)}{c\mu - \lambda} \tag{2.5}$$

To get μ^*, we find the value that minimizes the absolute distance between the theoretical W_q and the observed $\hat{W}_q$.

$$\min_{\mu} |W_q - \hat{W}_q| \tag{2.6}$$

Using a Python scalar optimizer, the optimum is $\mu^* = 1.97$ passengers per minute. We set this to be the service rate for regular passengers to $\mu_{reg} = 1.97$. For the PreCheck service rate, we set it to $\mu_{pre} = 1.97 * 1.33 = 2.262$. We multiply by 1.33 because a study found the PreCheck service to be 33% faster [12].

Our model design assumes the arrival rate for each passenger type is equal. We use a PreCheck probability (p_{pre}) to assign passenger types within the queue. p_{pre} is set to 0.27, an estimate given in a 2017 TSA report [12]. More recent sources put the number at 0.30 [8]. We use 0.27 because the more recent number is from a less reliable source. For the number of servers, we use data from Lockheed Martin, which suggests an average of 4.5 TSA agents, with a range of 2-12 across a day [2]. It is worth noting that the Lockheed dataset contains counts of the number of staff and the number of lines, not the number of TSA agents working at a security checkpoint. We assume half of the staff are allocated for security checkpoints while the others are allocated for other airport functions. Lastly, the value for total customers is given by the same LAX passenger data used to compute the arrival rate. The data puts the total number of passengers per day to be 34,000. [6].

3. Analysis

3.1. M/M/c Priority Queue. We run 500 simulations and average over each to measure the following metrics: average wait time, maximum wait time, and passenger throughput. Results are shown in Table 2. ρ values are included to highlight the stability condition that $\rho < 1$. This condition must be satisfied for both customer types.

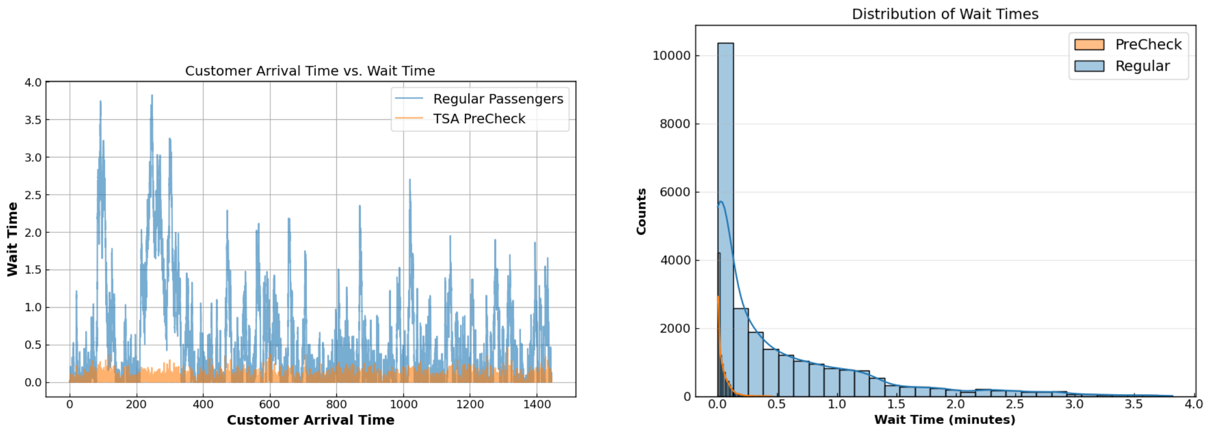
Table 1. M/M/c Priority Queue Metrics

Lane	**Avg Wait (min)**	**Max Wait (min)**	**Throughput (pass/min)**	ρ
PreCheck	0.04	0.75	6.37	0.75
Regular	0.53	9.79	17.22	0.998

The average wait time was lower for the PreCheck customers than for the Regular customers. This aligns with expectations because there are fewer PreCheck passengers, they receive priority, and PreCheck service times are quicker. The difference in maximum wait time between the groups is stark. This is due to the properties of the exponential distribution, which has a heavy tail (high variance, $\frac{1}{\mu^2}$) and is memoryless. Memoryless means the probability of waiting an additional time t is independent of how long you have already waited:

$$P(X > s + t | X > s) = P(X > t) \quad \forall s, t \geq 0.$$

Most customers experience short waits, but a small fraction experience extremely long waits. As $\rho \to 1$, $P(W_q > t)$ decays slowly, implying a higher chance of extreme wait times. Also worth mentioning is that the Regular group's ρ is extremely close to being unstable. Even though ρ_{reg} is close to one, the trace plots in Figure 2(A) show no signs of



(A) (B)

FIGURE 2. Trace Plots and Wait Time Distribution

divergence. In Figure 2(B), we see the distribution of wait times across passenger types. As we can see, the tails of the wait time distribution are much heavier for the regular group than for PreCheck.

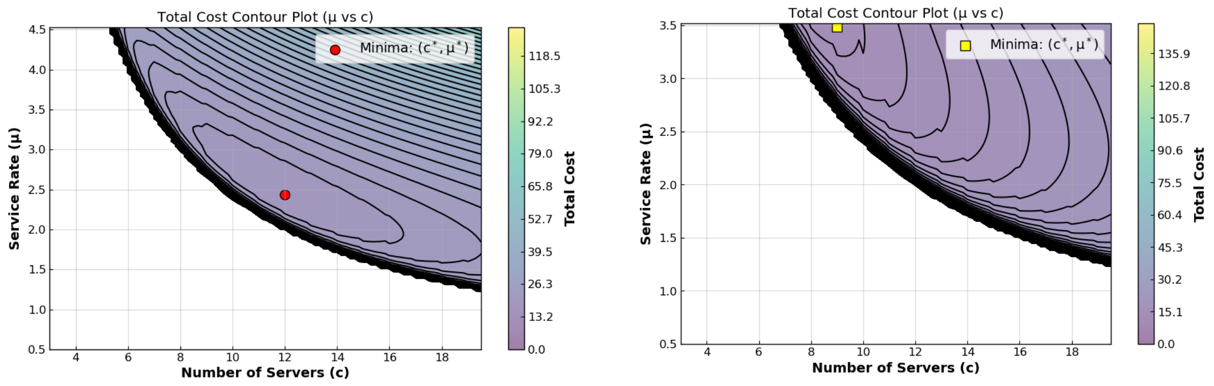
Now, we present an optimization problem, with a total cost objective and decision variables c and μ. There are three constraints with $\frac{\lambda}{c\mu} < 1$ as the one implicit constraint. The problem is formulated as follows:

$$\min_{c,\mu} T(c, \mu) = c_s c + (c_\mu \mu^3)c + c_w L(\lambda, c, \mu) \tag{3.1}$$

$$\text{s.t.}$$

$$\frac{\lambda}{c\mu} < 1;\ 3 \leq c \leq 15;\ 0.5 \leq \mu \leq 3.5$$

c_s is the cost of a server per hour, which can be interpreted as base pay. It is set to \$30 per hour based on the average TSA agent's pay [9]. c_w is the cost of a customer waiting per hour, which is set to \$37 per hour because it is the average wage in CA and reflects passengers' opportunity cost [3]. μ is the service rate. c_μ is the cost of a service rate per server per hour, which can be interpreted as bonus pay. We set it to be 6% of base pay, based on a \$2500 bonus on top of the provided base salary, adjusted to be in 2025 dollars [10]. L is the average number of passengers in the queue. We take the weighted average between the Regular and PreCheck average wait times to arrive at L. Lastly, c is the number of servers, which we keep at 12. We cube μ because additional bonus pay can only do so much to increase a server's service rate. The cubic form was chosen because it increases slowly at first, then increases rapidly, reflecting diminishing productivity. Humans have energy capacities and can't operate at high productivity levels for long periods. All units were converted to dollars per minute.



(A) (B)

FIGURE 3. Total Cost Loss Surfaces

Figure 3(A) displays the level sets of the total cost function, with cost increasing with the number of servers and the service rate. The minimum is achieved at $(\mu^* = 2.43, c^* = 12)$, with a total cost of \$18.18 dollars per minute. The optimal policy suggests that we should have 12 servers, each working at a rate of 2.4 people per minute, just under the PreCheck service rate. Since the minimum is at an interior point, this implies we should increase headcount or encourage more passengers to enroll in PreCheck . If the service rate was at the boundary, the policy would suggest decreasing headcount while maximizing each server's service rate by paying them larger bonuses. Figure 3(B) displays the optimal $(\mu^* c^*)$ when μ is squared rather than cubed (less diminishing returns in service rate).

3.2. **M/N/c Queue.** In the M/M/c model, the exponential distribution assumption for service time is strong because service time is arguably not exponential or memoryless. Here, we relax the assumption with an M/N/c queue. For tractability, we reduce the number of servers from 12 to 4. One server is devoted to Precheck, and the remaining three to regular screenings. Queues are parallel. The assumptions for the service rate are taken from section 2: $\mu_{reg} = 1.97$ and $\mu_{pre} = 2.262$. We were unable to identify statistics for the standard deviation of TSA security screenings, so we assume $\sigma_{reg} = 0.2$ and $\sigma_{pre} = 0.1$ minutes. For λ, we scale the prior value from section 3.2, since the number of servers was decreased. This results in $\frac{23.6}{3} = 7.87$ passengers. The PreCheck passenger proportion is calculated using a linear demand function defined in section 3.3, which gives a PreCheck proportion for a given input price.

With these elements defined, we determine the PreCheck price that corresponds with the minimum average wait time. We use prices from 0-120 dollars in increments of 10. 80 dollars was found to be the optimum. We then increase the granularity in the 70 to 90 dollar range to 0.25 dollar increments. This led to the surprise that for the 3:1 ratio of standard to PreCheck, the optimal price is \$76.75, the current PreCheck price. Using \$76.75 as the price point, we run 500 simulations and average results to get the average wait time, the maximum wait time, and the passenger throughput. Results are shown in Table 3.

The results in Table 2 conflict with the results from the M/M/c priority queue model. PreCheck passengers should have lower average wait times and maximum wait times. However, this is due to the server allocation. The result for throughput is intuitive because the regular screening has 3 lanes, the throughput per lane is lower than the PreCheck lane, $\frac{5.5}{3} = 1.8333$.

Under a fixed server allocation, getting to 70 percent TSA PreCheck is not a desirable objective, as it will lead to drastically increased wait times. To confirm, we calculated

Table 2. Queue Metrics Comparison

Queue Type	**Lane**	**Avg Wait (min)**	**Max Wait (min)**	**Throughput (pass/min)**
M/N/4	PreCheck (1 lane)	1.82	11.22	5.50
	Regular (3 lanes)	1.29	8.29	2.36
M/N/4	PreCheck (2 lanes)	0.61	4.96	4.53
	Regular (2 lanes)	0.76	6.18	3.33

that 70 percent participation in PreCheck would occur at a price of \$32.89. We ran the simulations again, in a 3:1 configuration. With this set-up, the average TSA Precheck wait time would be 2,381 minutes, or 1.65 days. This is unrealistic, as individuals would opt for the underutilized regular screening option.

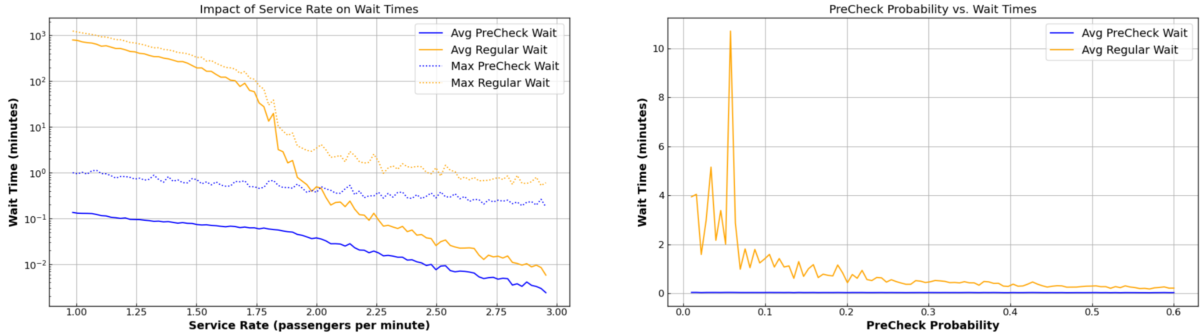
This raises an interesting question about the results if the server were split evenly between the two screening types. We modify the model so lanes are split evenly between the two screenings, keeping all other aspects of the model the same as prior. This returned an optimal price of \$46.50 and a participation rate of 57.6 percent. We then run 500 simulations at this price point and average to measure the average wait time, maximum wait time, and passenger throughput. Results are shown in Table 2. Precheck has a lower average wait time, a lower max wait time, and a higher throughput with equal lanes. This still does not meet the TSA's defined participation goal, but does provide additional confirmation that an increase in PreCheck enrollment and usage would lead to an improvement in passengers' wait times.

3.3. **TSA PreCheck Proportion Demand Function.** A key component of the M/N/c analysis is the demand function for the TSA PreCheck subscription. Our sources have the percentage of travelers using TSA PreCheck at 30 percent. The current price is \$76.75 for a 5-year membership [1]. For simplicity, we assume linearity, starting at 100 percent participation for a free membership. With that assumption, a cost of \$76.75, 30% of travelers are willing to pay, with 70% unwilling. Therefore, the proportion of PreCheck travelers changes by 76.75/70 = 1.096 per percentage change in participation. Therefore, the participation percentage, modeled as $\mathcal{P}$, at price p and r as the dollars cost per 1% drop in travelers:

$$\mathcal{P}(p) = \begin{cases} 1 & \text{if } p \leq 0 \\ 1 - \dfrac{p}{100r} & \text{if } 0 < p < 100r \\ 0 & \text{if } p \geq 100r \end{cases} \tag{3.2}$$

3.4. **Sensitivity Analysis.** We now consider the sensitivity of wait times to changes in two parameters for the priority queue model: service rate μ, and PreCheck probability (p_{pre}). For the M/N/c model, we analyze the sensitivity of the average wait time on the PreCheck price. We choose wait time are our primary result metric because wait time is important to both the passenger and the airport, while total cost is only important for the airport.

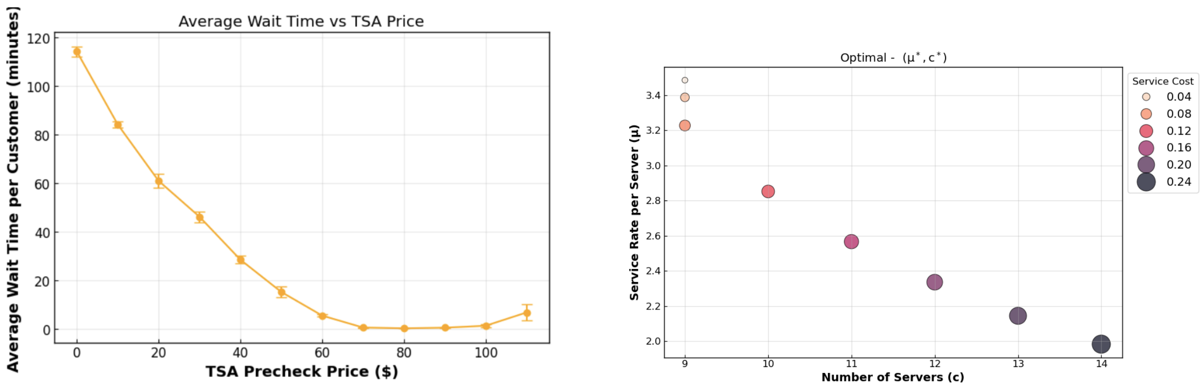
Focusing first on the M/M/c priority model, we examine how wait time changes as μ changes. The tested range is 50% to 150% of the value found through minimizing equation 2.6. From Figure 4(A), we see that wait time and μ are inversely related. As the service rate becomes quicker, the wait time decreases, an obvious conclusion. Increasing μ seems to matter more to regular than PreCheck passengers. Around $\mu = 1.9$, the average and



(A) (B)

FIGURE 4. Wait Time vs PreCheck Probability

max wait times drop significantly. This occurs when the stability condition is satisfied for Regular passengers ($\rho_{reg} < 1$). It is important that we test a range of service rates because service rates in reality are not fixed over time. Workers tire, so their service rates drop. It also differs by experience. More experience tends to coincide with quicker service. The analysis allows us to understand when workers should take breaks or have the next shift replace them, to keep queues stable. A sensitivity of wait time to λ, was omitted because it simply was the inverse of μ. As λ increases, wait times increase. A stability threshold was also present in this analysis.



(A) (B)

FIGURE 5. Wait Time and Optimal (μ^*, c^*) Sensitivity

We now examine the sensitivity of wait times to PreCheck probability (p_{pre}), Figure 4(B). As p_{pre} increases, the wait time for Regular decreases rapidly because fewer individuals are in the regular queue. Since PreCheck service times are quicker, there is no crowding-out effect; we do not observe an increase in the wait times for the PreCheck passengers. This highlights the reason why TSA wants as many fliers to be PreCheck as possible. Individuals are pre-screened, making service times quicker. Additionally, it alleviates the burden on Regular queues. A win-win situation.

For the sensitivity of the optimal (μ^*, c^*), shown in Figure 5(B), we observe that as the service cost increases linearly, the number of servers also increases linearly, and the service rate decreases linearly. The service cost reflects the strength of diminishing returns to paying servers a bonus for an increased service rate. If it is high, more money is needed to increase the service rate by the same percent. When diminishing returns are high, the airport hires more servers instead of paying performance bonuses, reflecting the tradeoff between maximizing service rate versus increasing headcount.

Lastly, we look at the sensitivity of the M/N/c model. In Figure 5(A), each point denotes the average wait time for the model, averaged over five simulations. The wait time and price are inversely related, because if the price is low, everyone crowds to the PreCheck line, and wait times go up. As price increases, people move from the PreCheck line to the regular line, lowering wait time until it begins to tick up again as individuals crowd the regular line. Note, the crowding-out effect is a result of fixed server allocation, which is not present in the M/M/c priority model.

4. Discussion

To model heterogeneous passenger types, many research papers opted for M/M/c with parallel queues, a dedicated line for each type. The advantage is that the model has closed-form solutions. Expected wait times, stability conditions can be described with equations. In this paper, we opted for more complex models with no closed-form solutions and relied on simulations to approximate key metrics.

The M/M/c priority queue model highlights the criticality of $\rho < \frac{\lambda}{c\mu}$. If the condition isn't satisfied, the system blows up. Therefore, an airport must select c and μ carefully. The sensitivity analysis illustrates the regimes for which a queue is stable versus unstable. The model also illustrates the effect of PreCheck probability on wait time, showing no evidence of crowding out. As the number of PreCheck passengers increases, both passenger types benefit. The productivity gain comes because the PreCheck service rates are quicker due to pre-screening.

In our M/N/c model, we scaled down the model to a third of the M/M/c model, for tractability. The results of the M/N/c show that wait times for PreCheck are much longer than for regular passengers. This reflects the inefficiency of a parallel queue with fixed server allocations. The model showed crowding out effects. The distribution of wait times differs from the M/M/c queue in that they are more uniformly distributed. We also created a demand function and estimated the optimal TSA price. This contrasts with the M/M/c optimization, because here we focused on the perspective of the consumer.

In terms of parameter values, many papers did not provide units or omitted values completely. We relied on data and calculated the parameter values ourselves. Some parameter values, like arrival rate, were easy to compute because passenger counts were available. However, the service rate was not. We relied on the assumption that data was generated by an M/M/c queue to back out the service rate. A parallel M/M/c priority queue model was also attempted, but it did not satisfy the stability condition. Therefore, we did not pursue this design further. As for the total cost optimization, we framed it from the perspective of the airport and the consumer. When server allocation is flexible, the best option is to have every passenger be PreCheck. However, when allocation is fixed, it is best that only a proportion of passengers are PreCheck.

5. Conclusion

Regardless of whether one is traveling for leisure or business, queuing up for security is often a hurdle. Airports are tasked with striking the right balance to ensure adequate wait times and throughput while minimizing cost and avoiding overwork. To model such a system, we extended the standard, parallel M/M/c model to an M/M/c model that uses a priority queue to handle two types of customers, PreCheck and Regular. We then relax distributional assumptions by introducing an M/N/c model. Our priority queue model highlights the win-win benefit of converting passengers to PreCheck. As the PreCheck proportion grows, wait times for everyone go down. In contrast, our M/N/c model showed the inefficiency and crowding out due to fixed server allocation. We examined the optimal policy of server and service rates for a given arrival rate, from the perspective of the airport. From this analysis, we gained insight into how to allocate servers by number and experience for busy versus slow times. We also framed cost in the perspective of the consumer, showing the optimal TSA PreCheck using our M/N/c model. The strength of our approach is that parameter values are estimated from data, making them flexible. If future research wants to extend our model to other airports, they can use our approach to estimate parameter values. Analysis on other metrics besides average wait time remains as a possible extension of our model.

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6. Statement of Individual Contribution

Kaylee

- Wrote the first draft of the report and proofread. Worked on M/M/c and optimization based on airport perspective, equation (3.1).
- Parameter research and parameter estimation via equations 2.2 - 2.6.

Tadhg

- Built out the single checkpoint simulation using a normalized service time and evaluated in the 3:1 configuration and 2:2 configuration
- Wrote the first draft analysis on the M/N/C models.
- Wrote the first iteration of the powerpoint and combined the notebooks